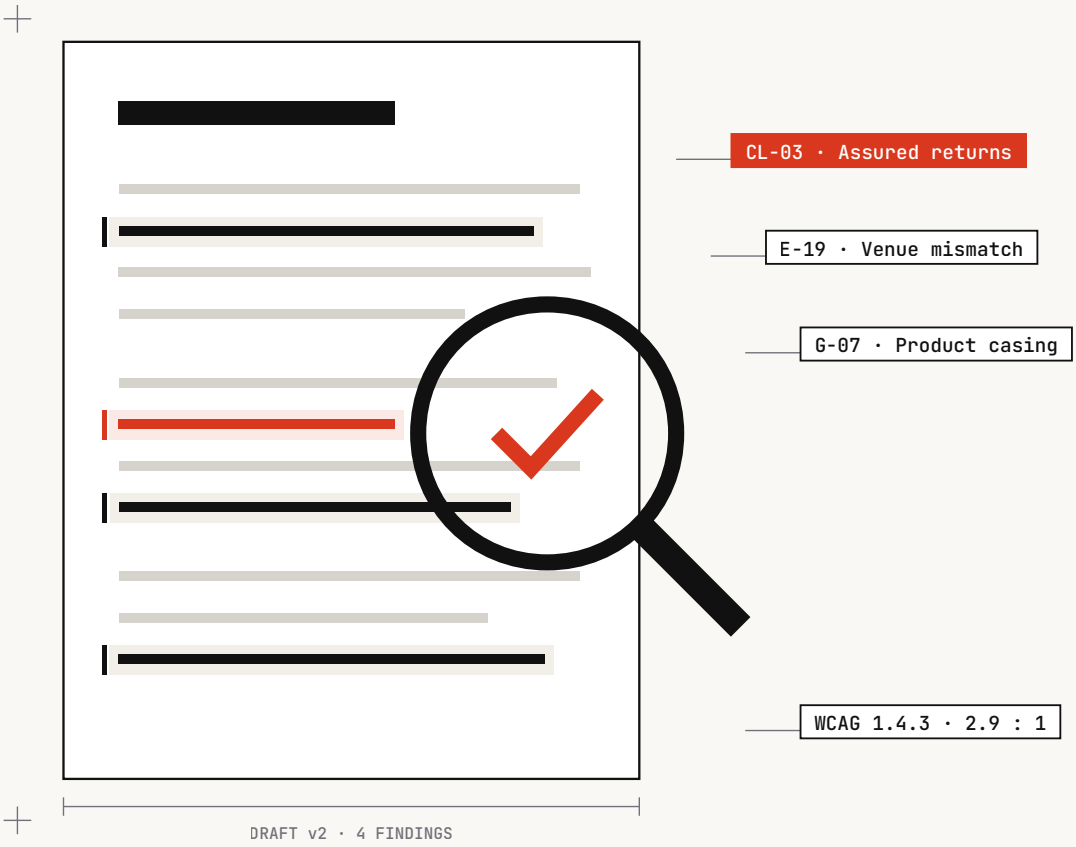
ROUND 1 SUBMISSION · THE PLAN

BrandLens

Pre-publication brand governance. Every draft is checked against an organisation's approved rules, facts and assets, and every flag cites the source it was checked against.



PROBLEM STATEMENT

Inkloom AI — Intelligent brand-governance platform

TEAM

PENDING · team name

Maanit Gupta · Jagpreet Singh
SRM Institute of Science and Technology

SUBMITTED

PENDING · submission date

Contents & Requirements Map

Each of the committee's 11 required items, and where it is answered in this document.

#	REQUIRED ITEM	WHERE IT IS ANSWERED	PAGE
01	**Understanding of the problem statement**	Evidence & our reading of the brief; root cause	p. 4-5
02	**Users affected**	Users & primary discovery; current vs target workflow	p. 6-7
03	**Proposed solution**	Executive summary; proposed solution; verification pipeline	p. 3, 8, 12
04	**Complete product workflow**	End-to-end workflow; user journey storyboard; state machine	p. 9-11
05	**Main features**	MoSCoW scope & check matrix	p. 13
06	**Technology stack**	Architecture; hardest problem & stack rationale; schema	p. 14-16
07	**Each member's responsibilities**	Build plan & roles; team details	p. 20, 26
08	**Development plan across all three rounds**	14-day build plan; plan across three rounds	p. 20-21
09	**Expected impact**	Metrics, baselines & measurement	p. 24
10	**Future scope after the event**	Roadmap & business model	p. 25
11	**Complete team and participant details**	Team & participant details	p. 26

Contents

01 Cover
02 Contents & requirements map
03 Executive summary
04 Evidence & reading of the brief
05 Root-cause analysis
06 Users & primary discovery
07 Current vs target workflow
08 Proposed solution
09 End-to-end workflow
10 User journey storyboard
11 State machine & permissions
12 Verification pipeline
13 Features & scope
14 System architecture
15 Hardest problem & stack rationale
16 Schema & data sources
17 Originality
18 Responsible design
19 Governance insights
20 14-day build plan & roles
21 Plan across three rounds
22 Feasibility, risk & cut-lines
23 Validation & research evidence
24 Expected impact
25 Future scope
26 Team & participant details
27 References & disclosures

Executive Summary & Verification Architecture

Problem

Content is produced by employees, agencies and LLM tools in parallel. Incorrect claims, cross-campaign fact errors, tone drift and off-brand visuals pass a manual review chain that runs 3–4 rounds and still depends on whoever checks last.

Solution

BrandLens runs each draft through a 6-stage pipeline: deterministic filters, pgvector retrieval over approved rules and facts, a structured LLM pass, and a citation gate. Findings show the rule or fact ID, an explanation and an intent-checked rewrite. Publishing always requires a human sign-off.

Target users

Organisations with distributed content production: creators and agencies, approvers, legal/compliance and accessibility reviewers, and the brand owner who maintains the guidelines.

77%

of respondents see off-brand content created at their company (2021)

Marq (formerly Lucidpress), 2021 (n=452)

98%

of ads ASCI scrutinised in FY 2024–25 needed some modification

ASCI Annual Complaints Report

70%+

of CMOs are concerned about GenAI's impact on creativity and brand voice

BCG CMO survey, 2024 (n=200)

79.1%

of top home pages have low-contrast text

WebAIM Million, 2025

CORE ARCHITECTURAL BETS

B1 · Citation gate. An AI finding is displayed only if it references an existing approved rule or fact ID.

B2 · Intent-verified rewrites. Fixes are conditioned on stated goal, audience and CTA, then verified locally in <15 ms.

B3 · Guideline telemetry. Dismissals and overrides produce a per-rule Guideline Clarity Score.

Primary Discovery & Verification Snapshot

Four discovery interviews, October 2026, and the architectural bet each one shaped.

INTERVIEWEE	ENVIRONMENT	FAILURE OBSERVED	BET IT SHAPED
Pallavi Biala Marketing, Leo Club, Panjab University	University service org, Chandigarh	LLM pulled venue and timing from a parallel event; post went out with the wrong details	B1 Fact Vault cross-check of dates, venues and campaign facts
Harshit Singh Raghuvanshi Founder, marketing agency	Agency, Mohali	Work built for the creator's taste, not the audience; rewrites land at founder sign-off	B2 Mandatory audience/intent framing; risk-sorted queue
Shaurya Sandal Marketing, Cultural Society, Symbiosis	Student cultural org, Pune	Late, vague feedback that contradicts the previous round	B1 B2 Rule-cited fixes before submission
Devansh Content creator for small businesses	SMB content, multi-client	Static PDF style guide; no visibility into which rules confuse creators	B3 Rule cards from PDFs + Guideline Clarity Score

Industry & Regulatory Evidence: Why Manual Review Fails at Scale

Content volume is rising, a growing share is LLM-generated, and regulators are scrutinising promotional claims more closely. Review processes sized for a small in-house team do not scale with these three trends.

85%

of respondents saw demand for content rise in the previous year

Marq (formerly Lucidpress), 2021

81%

of organisations reported dealing with off-brand content (2019)

Lucidpress State of Brand Consistency, 2019

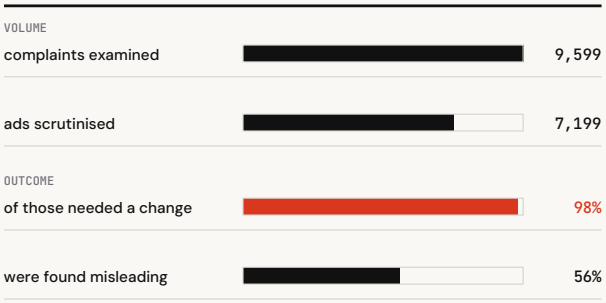
10–20%

revenue lift reported by 68% of respondents when brand content stays consistent

Marq (formerly Lucidpress), 2021

Unsubstantiated claims are a regulatory exposure

ASCI (Advertising Standards Council of India), FY 2024–25

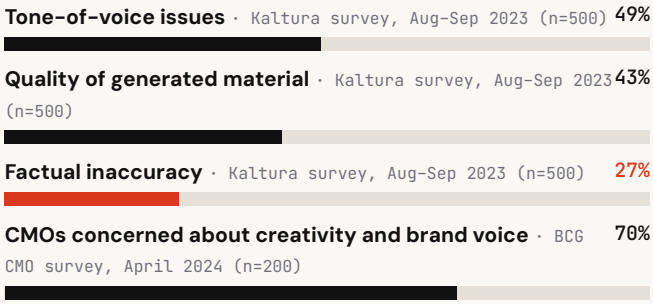


ASCI reported a **70% rise** in complaints for April–September 2025 vs the same period in 2024. Unsubstantiated and exaggerated claims are a pattern-matchable category that a pre-publication check can detect.

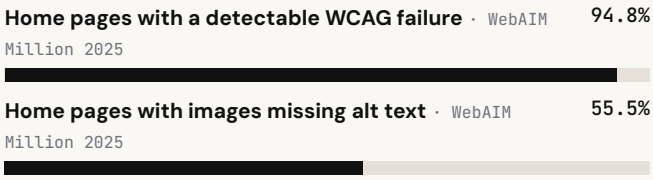
Sources: ASCI Annual Complaints Report 2024-25 press release; ASCI Half-Yearly Complaints Report 2025-26.

GenAI increases output and review load

Top concerns marketers report about AI-created content



Accessibility failures are common and measurable

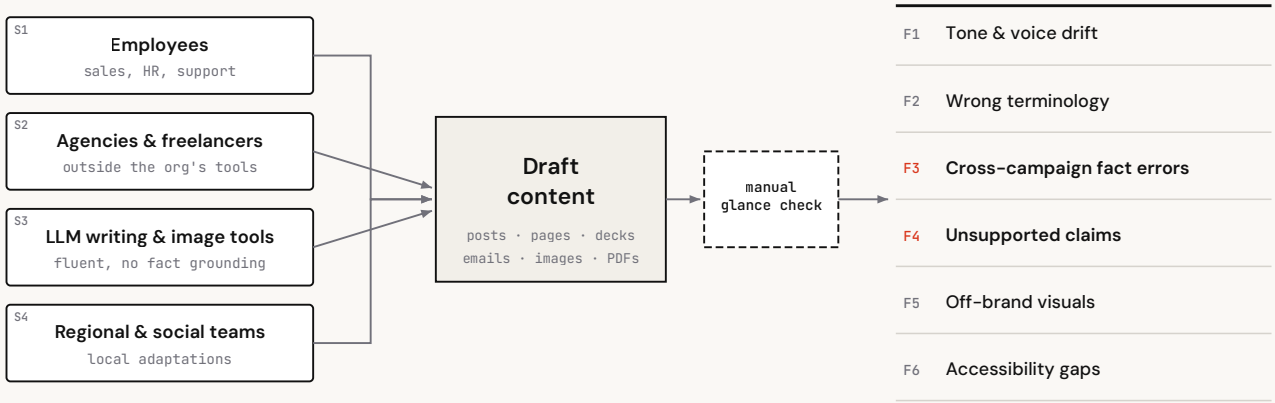


Our reading of the brief

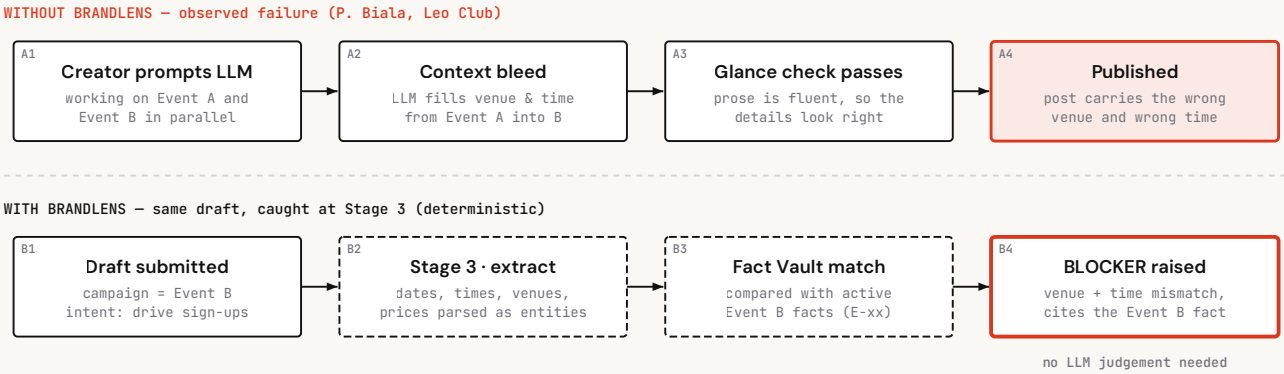
The part we solve	**BrandLens verifies claims and campaign facts before human review; tone, terms and accessibility checks support that; visual checks follow in Round 2.** Reviewers read wording but no round checks facts against a source of truth, so that is where we start. Priority follows the evidence: claims, then tone, then accessibility.
Out of scope	Post-publication monitoring, content generation, digital asset management.
Tensions	Automation vs human control · scale vs explainability · AI use vs confidentiality.

Root-Cause Analysis: Multi-Source Content Drift & LLM Context Bleed

Drafts come from several contributor types, each with partial access to the guidelines and current facts. They converge on a manual check that inspects wording but does not systematically verify facts against a source of truth.



Real Failure Trace: LLM Context-Bleed & Fact Drift



Why fluent LLM copy defeats manual checks

Pallavi was running two events at once. The LLM merged their context and produced a post with the wrong location and timing. The copy read well, so she assumed the details were correct and submitted it. Reviewers in a 3–4 round chain read for tone and wording; none of the rounds compares dates and venues against a canonical record.

Contributing causes

- Guidelines live in a static PDF that is rarely re-opened
- Facts (dates, fees, venues) change; old values persist in templates and prompts
- Reviewers interpret "on-brand" differently, so feedback conflicts between rounds
- Agencies cannot see internal source material

Users & Primary Discovery: Four Content Environments

We interviewed four people in October 2026 across an agency, two student organisations and small-business content work. Each interview covered their approval chain, a recent failure, and what a reviewing tool would need to do for them.

[01 // P. BIALA]

Pallavi Biala

Marketing, Leo Club
Panjab University, Chandigarh

WORKFLOW REALITY

Every post goes through 3–4 rounds of approval and often stalls inside the same hierarchy.

FAILURE REPORTED

While handling two events, her LLM pulled the venue and timing from the wrong project. The fluent copy hid the error and the post was submitted with both details wrong.

REQUIREMENT FOR BRANDLENS

Automatic cross-check of dates, venues and campaign facts against an approved Fact Vault before human review.

[02 // H. RAGHUVANSHI]

Harshit Singh Raghuvanshi

Founder, marketing agency
Mohali, Punjab

WORKFLOW REALITY

3-stage manual chain: creator/ designer → senior peer check → founder sign-off.

FAILURE REPORTED

“The dumbest mistakes are not knowing what the viewer wants to view — you just don't make everything for yourself.” Misaligned work is rewritten at founder level.

REQUIREMENT FOR BRANDLENS

Mandatory audience & intent framing at submission and a risk-sorted queue, so he reviews alignment and diffs, not every line.

[03 // S. SANDAL]

Shaurya Sandal

Marketing, Cultural Society
Symbiosis, Pune

WORKFLOW REALITY

High-velocity event and sponsor campaigns; posts and pages ship under tight deadlines.

FAILURE REPORTED

Feedback arrives days late, is vague (“make it more on-brand”) and contradicts the previous round, because reviewers read the guidelines differently.

REQUIREMENT FOR BRANDLENS

Rule-cited fixes in seconds before submitting, with rewrites that keep the original campaign hook.

[04 // DEVANSH]

Devansh

Content creator
Small-business clients

WORKFLOW REALITY

Maintains brand identity across several clients and channels at once.

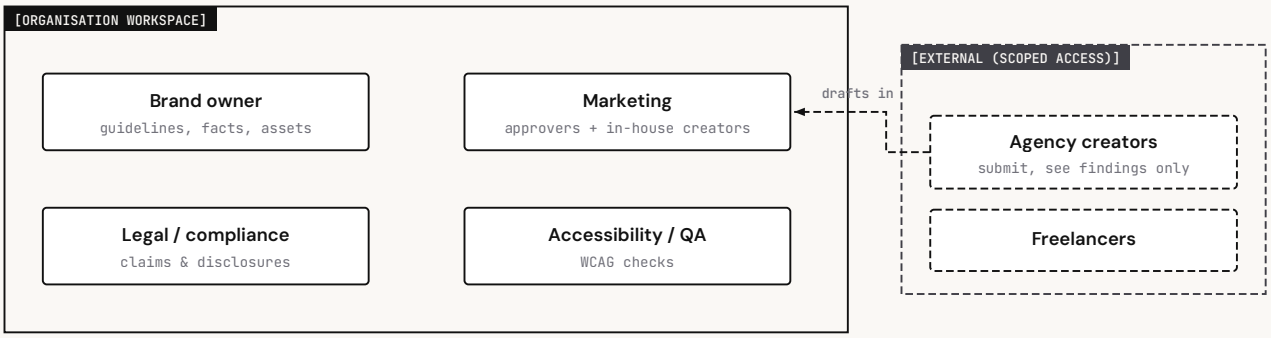
FAILURE REPORTED

Owners demand strict consistency but supply a static PDF style guide that nobody re-reads, with no signal on which rules confuse creators.

REQUIREMENT FOR BRANDLENS

Machine-readable rule cards from PDFs, plus telemetry on which rules trigger repeat mistakes or dismissals.

Organisational setting & roles in the product



Channels

Social posts, event pages, web copy, email campaigns, sales decks (PDF), banners and product images.

Demo scenario

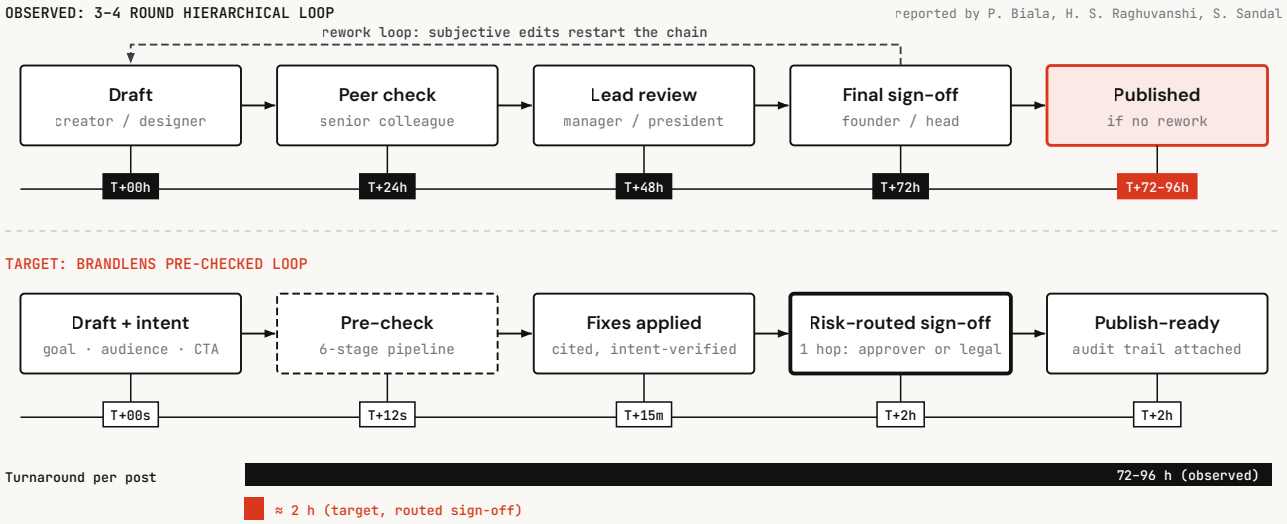
A fictional Indian fintech brand, “PayWise”, running a festive SIP campaign and a webinar. Chosen because financial claims carry the highest regulatory risk and the campaign has many dated facts.

Adoption trigger

A published correction, a regulator notice, a rebrand, or rolling out LLM tools to the content team.

Workflow Benchmark: 4-Round Hierarchical Bottleneck vs. Pre-Verified Routing

Three of four interviewees described a sequential approval chain in which every round re-reads the full draft. BrandLens moves verification to submission time and sends each draft to one reviewer chosen by risk.



Finding 1 · Feedback without a rule reference repeats rounds

Shaurya reported edits that contradict the previous round. Attaching a rule ID to each comment gives reviewers a shared reference.

Finding 2 · Senior reviewers line-edit

Harshit's founder-level sign-off catches audience misalignment that earlier rounds miss. A diff and risk score let him review intent, not every line.

Finding 3 · No round checks facts

Pallavi's post passed several hands with a wrong venue and time. Fact verification has to be automatic and deterministic.

Observed timings are the turnaround interviewees described for a typical post. BrandLens timings are design targets that will be measured in the Round 2 benchmark (page 23).

Proposed Solution: Source-Cited Review Across Six Check Dimensions

The brand owner converts the style guide into approved rule cards, facts and assets. Each draft (text, document or image) is checked on six dimensions. A finding includes the matched rule or fact ID, an explanation and an intent-verified rewrite. Approval and publishing decisions stay with people, and each decision is logged.

BrandLens · 6 check dimensions

01 / TONE
Tone & voice
formality, warmth, brand personality vs. guide

02 / TERMS
Terminology
preferred / banned terms, product names, casing

03 / FACTS
Messaging & facts
dates, venues, fees vs. the Fact Vault

04 / CLAIMS
Claims
superlatives, guarantees, needs evidence or disclosure

05 / VISUAL
Visual consistency
logo use, palette, fonts, imagery rules

06 / ACCESS
Accessibility
contrast, alt text, reading level, link text

Value per user role

ROLE	CURRENT COST	WHAT BRANDLENS PROVIDES
Creator	Late, vague, contradictory feedback across 3–4 rounds	Rule-cited fixes in seconds before submission, conditioned on the stated campaign intent
Approver	Line-by-line reading of every draft; 72–96 h per post	Risk-sorted queue, version diff and brand-fit score
Legal	Claims discovered after publication, or not routed at all	Claims extracted and routed with the matching fact, or a note that none exists
Brand owner	Static PDF guide, no data on which rules confuse people	Versioned rule cards and a per-rule Guideline Clarity Score

System Guarantees & SLA

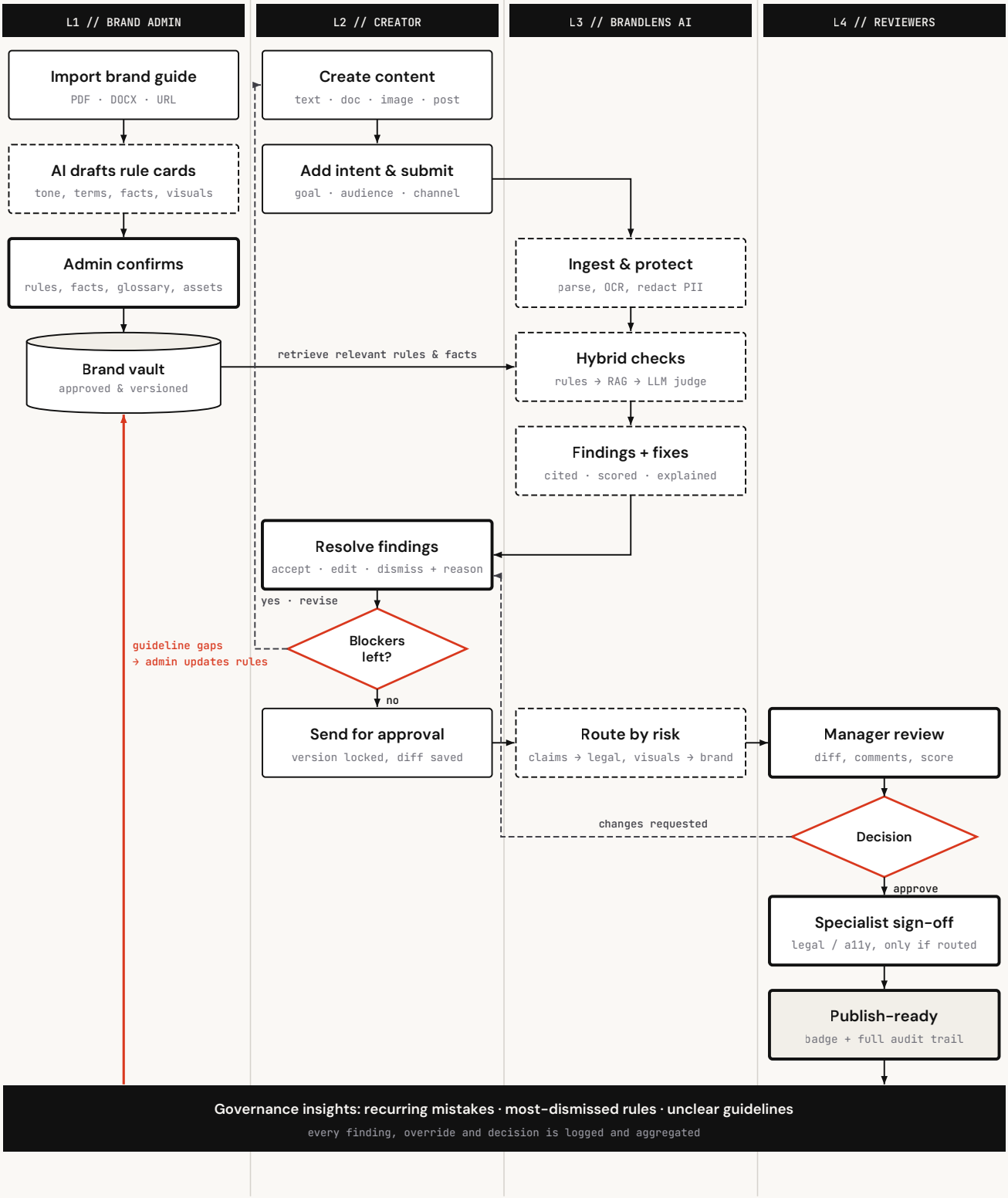
100%
RULE-CITED FLAGS
citation gate drops any finding without a valid rule or fact ID

0
UNAPPROVED AUTO-PUBLISHES
publish-ready state requires a recorded human approval

<15 s
P95 TEXT ANALYSIS
drafts up to 2,000 words, end to end

<15 ms
INTENT VERIFICATION
local embedding similarity + CTA retention per rewrite

End-to-End Workflow: Guideline Ingestion to Publish-Ready Sign-off



AI step
Automated; produces suggestions only.

Human checkpoint
Thick border = a person must decide.

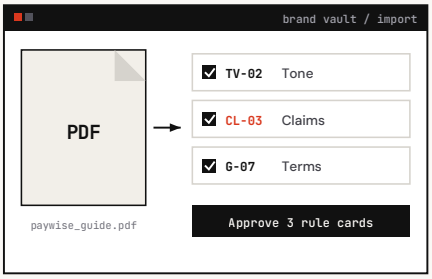
Decision
Blocking findings or reviewer choice.

Feedback loop
Insights flow back into the guidelines.

User Journey Storyboard: One Campaign Post, Six Frames

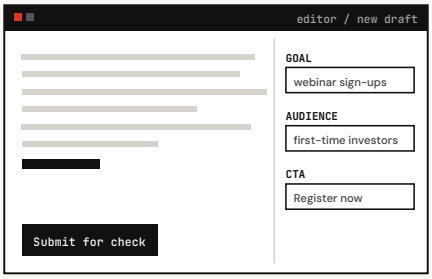
The PayWise Diwali SIP caption from the demo scenario, followed from guideline import to a rule rewrite. Each frame is a screen in the product.

[F01 // SET UP]



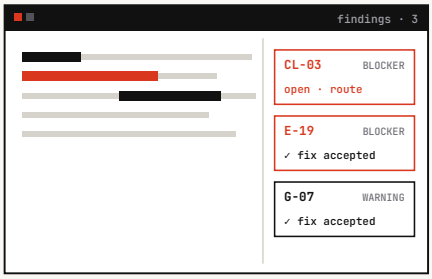
The brand admin uploads the PayWise guide; AI drafts rule cards; the admin approves them.

[F02 // DRAFT]



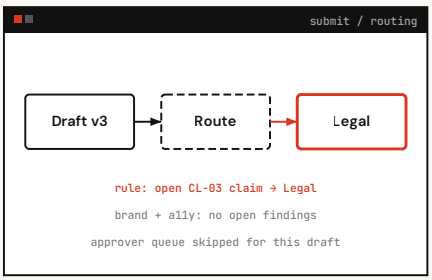
A creator writes the Diwali SIP caption and states its goal, audience and CTA.

[F03 // PRE-CHECK]



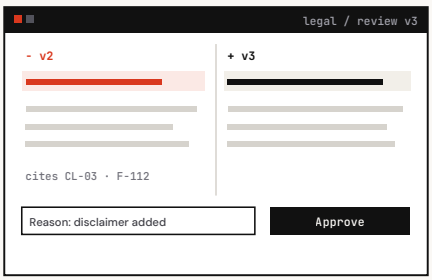
Findings appear: CL-03 blocker, E-19 fact drift, G-07 casing; the creator accepts two fixes.

[F04 // ROUTE]



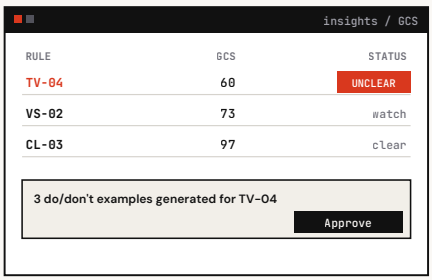
Submitting routes the draft to Legal because a claim remains.

[F05 // REVIEW]



Legal reviews the diff and citations, then approves with a logged reason.

[F06 // LEARN]

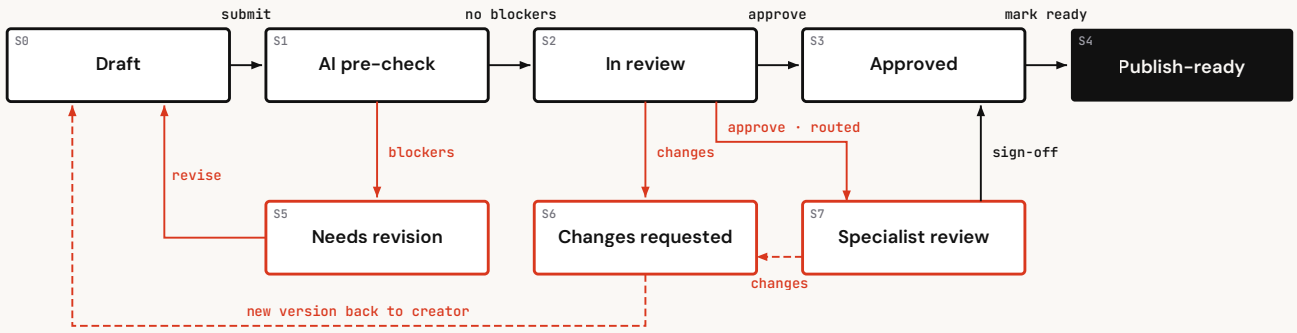


The brand owner sees TV-04 flagged UNCLEAR and approves new examples.

Wireframes are illustrative; the interface will be built during the build window. Rule and fact IDs match the finding mock-up on page 12.

Content State Machine & Role-Based Permissions

Each content item has one state and one owner at any time. Every transition is written to the audit log with actor, timestamp and reason; the same log feeds the Guideline Clarity Score.



Any state can be archived by a manager or admin; archived content keeps its full history. BrandLens never publishes: a person marks content Publish-ready after the required sign-offs, then posts it in their own channel.

Role-based permissions

ACTION	AGENCY CREATOR	CREATOR	MANAGER	SPECIALIST	BRAND ADMIN
Define guidelines, facts, glossary, assets	○	○	○	○	●
Submit content & run AI pre-check	●	●	●	○	●
See full finding + cited source text	◐	●	●	●	●
Accept / edit / dismiss a finding	●	●	◐	◐	●
Comment on a version	●	●	●	●	●
Approve or request changes	○	○	●	●	●
Sign off claims / accessibility	○	○	○	●	◐
View insights dashboard	○	○	◐	○	●
View audit log	○	○	●	●	●

● allowed ◐ limited (agencies see the rule summary, not confidential source text; specialists act only on findings routed to them) ○ not allowed

Collaboration features

Threaded comments

Anchored to a sentence, a finding or an image region; @mention reviewers.

Versions & diff

Each submission is immutable; reviewers see what changed and which findings it resolved.

Dismiss with reason

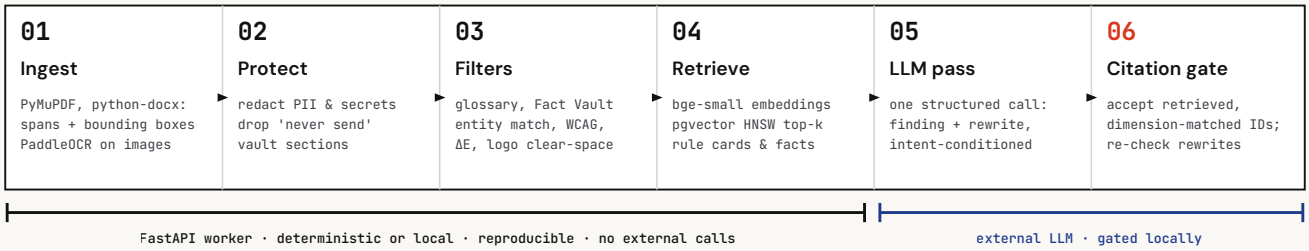
“Approved exception”, “doesn't apply”, “AI is wrong” — reasons feed guideline-gap insights.

Override with rationale

Reviewers can approve with open warnings; the decision and rationale are logged.

6-Stage Verification Pipeline: Deterministic Filters, pgvector RAG & Citation Gate

Stages 1–4 run inside the FastAPI worker with no external calls. The LLM is invoked once per draft for judgement-based dimensions, and stage 6 accepts a citation only if that rule or fact ID was retrieved for this draft and its dimension matches the finding.



Finding output (product mock-up)

Instagram caption · Diwali SIP campaign · v2 · LLM-assisted draft · intent: webinar sign-ups

> "This Diwali, start a festive SIP on Paywise app with 0% exit load and lock in 14.2% assured annual returns — webinar registration closes Oct 18 at Hall B, Chandigarh!"

`BLOCKER · Unsupported regulatory claim` `confidence 0.96`

"14.2% assured annual returns" presents a historical figure as a guarantee on a market-linked product, against ASCI/SEBI norms.

Cites: `CL-03` No guaranteed/assured return language on market-linked products · `F-112` Flexi-Cap 3-yr historical CAGR is 14.2% as of Q3 2026; requires the disclaimer "Mutual Fund investments are subject to market risks, read all scheme related documents carefully."

> `Fix:` "…explore our Flexi-Cap fund (3-yr historical CAGR 14.2%, Q3 2026). Mutual Fund investments are subject to market risks, read all scheme related documents carefully."

`BLOCKER · Cross-campaign fact drift` `deterministic match`

"0% exit load" and "Oct 18 at Hall B" conflict with active campaign facts.

Cites: `F-048` 1% exit load on redemptions within 365 days · `E-19` Diwali Wealth Webinar: online only, Oct 28, 7:00 PM IST

> `Fix:` "…register for our online Diwali Wealth Webinar on Oct 28 at 7:00 PM IST." (exit-load claim removed)

`WARNING · Terminology` `glossary G-07`

"Paywise app" → **"PayWise"** (camel-cased, without the word "app").

`✓ Accept` `✎ Edit` `× Dismiss with reason` `→ Route to Legal`

Finding schema

span	LLM returns the exact quote; offsets, or `(page, x0, y0, x1, y1)` for PDFs and images, are computed locally; a quote not found drops the finding
cites	≥ 1 rule or fact ID from this draft's retrieved top-k, dimension-matched
severity	blocker · warning · suggestion
confidence	deterministic = 1.0; LLM = retrieval similarity × agreement of two samples; < 0.6 routes to a human
source	deterministic or llm
rewrite	optional; intent-verified

Intent verification in <15 ms

The LLM returns the finding and its rewrite in a single structured-output call, conditioned on the creator's stated goal, audience and CTA. The worker then checks each rewrite locally:

- `bge-small-en-v1.5` cosine similarity to the original ≥ **0.82**, a threshold calibrated on the golden set
- CTA and campaign keywords retained
- the rewrite is re-run through Stage 3 and the cited rule, so a fix cannot break a rule
- fail → rewrite dropped; finding shown without a fix

MoSCoW Feature Scope & Multi-Format Check Matrix

Scope is ordered so the full review loop works on text first. The "Must" column covers every capability in the problem statement for text content; the remaining columns add formats and depth.

MUST · core loop

- Brand vault: PDF guide import, AI-drafted rule cards, admin confirmation
- Fact Vault (campaign, product, event facts), glossary, asset library
- Text review: tone, terms, facts, claims, readability
- Cited findings with explanation and fix
- Accept / edit / dismiss with reason
- States, comments, versions, approve / reject
- Role-based access via Postgres RLS
- AI-use disclosure and audit log

SHOULD · differentiators

- Intent framing at submission + local intent verification
- Image checks: OpenCV logo and clear-space, OCR contrast, ΔE on fills
- Risk-based routing to legal / accessibility
- Guideline Clarity Score dashboard
- Version diff showing resolved findings
- Slack / email approval notifications

COULD · polish

- PDF deck review with page thumbnails and bbox overlays
- Regulatory rule packs (ASCI-style claim rules)
- Brand-fit score trend per team
- Bulk review of a campaign
- Hindi and Tamil tone checks (multilingual support deliberately deferred)

WON'T · this event

- Auto-publishing to social channels
- Video and audio analysis
- Fine-tuning models on client data
- Native mobile apps
- Billing and multi-tenant admin

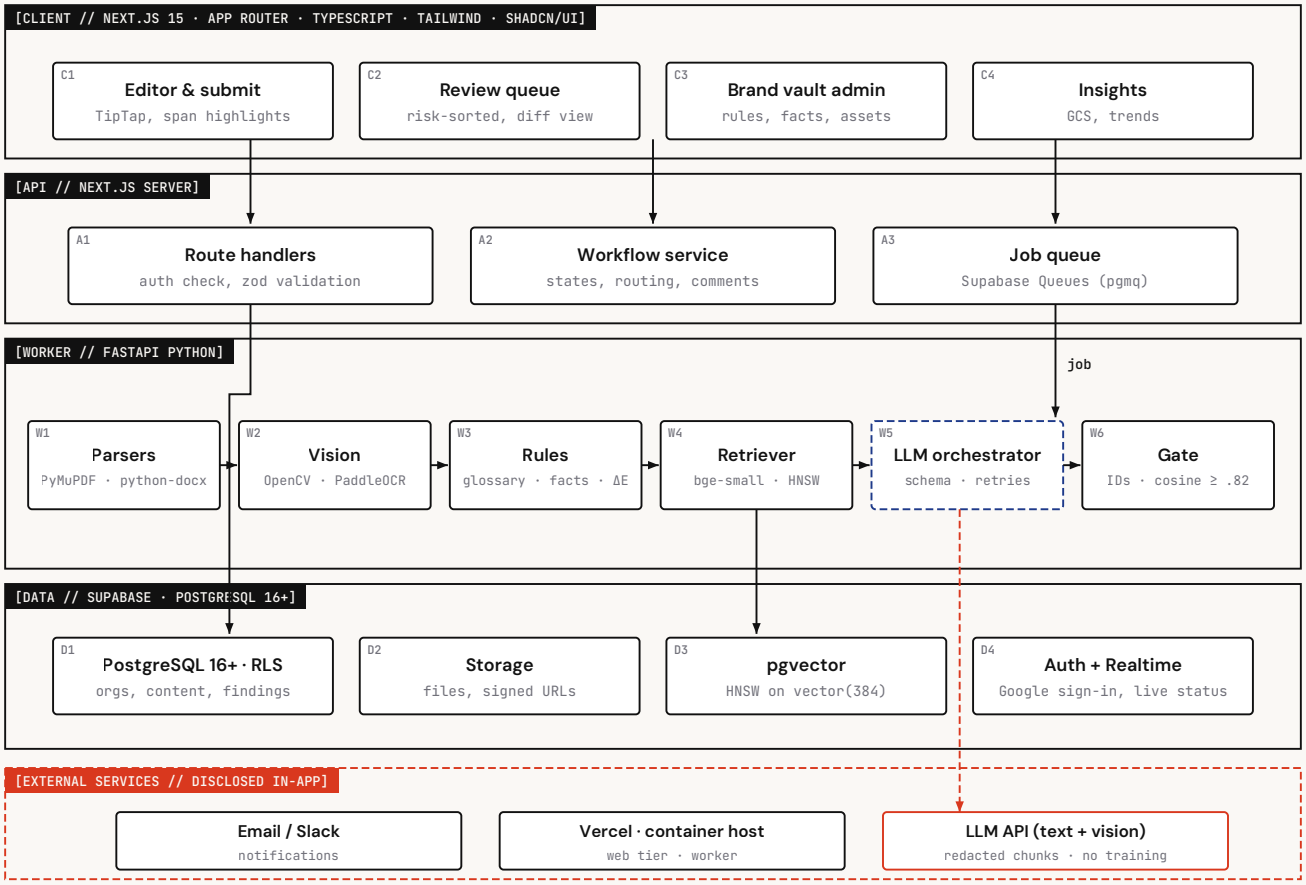
Supported content formats

Plain text / Markdown	MVP
Web page (URL / HTML)	MVP
Word (.docx)	MVP
Images (PNG, JPG)	MVP
PDF decks & brochures	Round 2
Social post = copy + image	Round 2
Email HTML	Later
Video captions / scripts	Later

Check matrix by dimension

DIM.	DETERMINISTIC (WORKER)	LLM (CITED)
Tone	exclamation density, caps ratio, banned phrasing	voice match vs. rule examples
Terms	glossary + casing, inflection-aware Aho-Corasick	is the token the product name?
Facts	dates, venues, fees, rates extracted and matched to Fact Vault; expiry	paraphrased or outdated facts
Claims	superlative and guarantee lexicon	claim extraction + evidence match
Visual	OpenCV ORB + matchTemplate logo bbox; pixel variance in 0.25× clear-space; CIEDE2000 ΔE on segmented fills and type	imagery style vs. guide
Access.	OCR glyph vs. background-ring luminance (WCAG 2.2 AA); reading grade; link text	alt-text quality

System Architecture: Next.js 15 Client, FastAPI Verification Worker & Supabase RLS



Key technical decisions

Separate Python worker

PDF parsing, OpenCV, OCR and local embeddings are Python-native. Running them in a FastAPI worker keeps heavy compute off the web tier and lets it scale on its own. Jobs arrive through pgmq, so no extra broker is needed.

Deterministic first, one LLM call

Terms, facts, contrast and logo checks resolve without a model. The LLM gets one structured-output call per draft for tone, messaging and claims, returning findings and rewrites together. This bounds latency and cost.

Postgres as the backbone

Supabase provides auth, row-level security, pgvector, storage, realtime and queues in one database. Tenant isolation is enforced by RLS policies, not application code.

Hardest Technical Problem & Stack Rationale

[THE PROBLEM]

Turning a model's judgement into a finding that is grounded, located exactly and rarely wrong.

Why it is hard

Models paraphrase, cite IDs that don't fit, miscount character offsets and overstate their own confidence. Any one of these turns a useful reviewer into noise that creators learn to ignore.

Fallback

Any dimension that scores under 85% precision on the golden set ships as suggestions only, never as blockers. **Second challenge:** extracting rule cards from free-form PDF guides, handled by a fixed rule-card schema and admin confirmation of every card.

How the architecture answers it

01	02	03	04	05	06
Deterministic checks first	**Campaign-scoped** retrieval	**Quote-anchored** spans	**Retrieved-ID** citation gate	**Re-checked** rewrites	**Precision gate** in CI

Deterministic checks resolve what they can before any model call; retrieval is scoped to the draft's campaign; the LLM returns exact quotes and offsets are computed locally; a citation is accepted only if it was retrieved for this draft with a matching dimension; every rewrite is re-run through the checks; CI blocks any prompt or model change that drops golden-set precision.

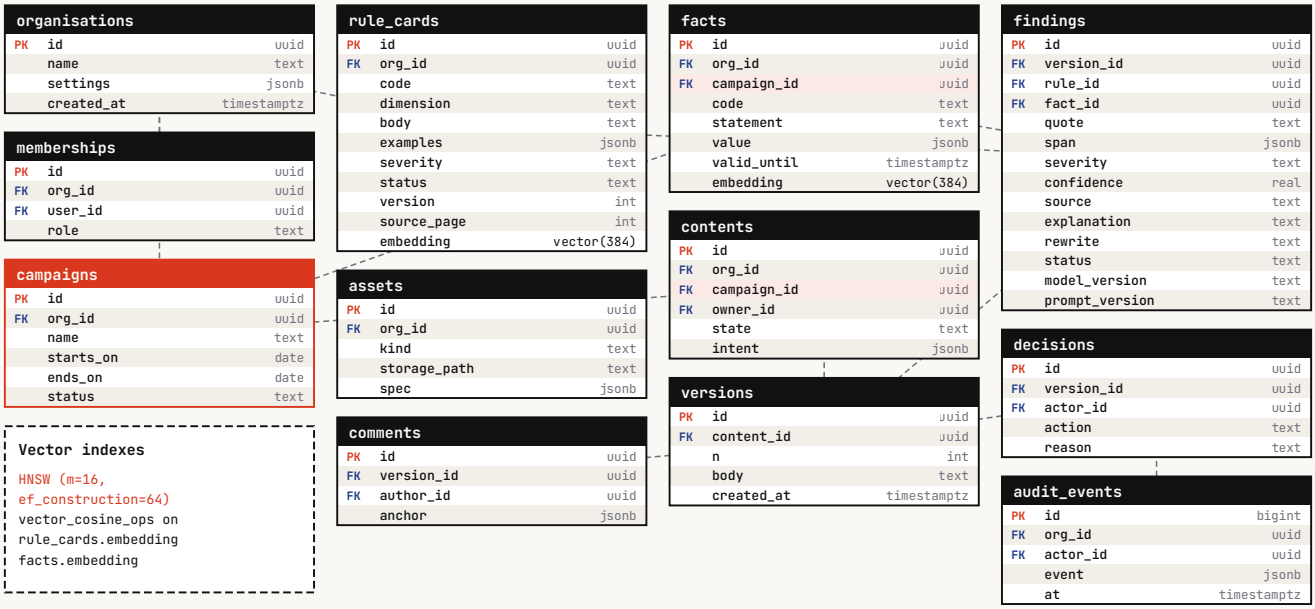
Why this stack

LAYER	CHOICE	REASON	REJECTED
Frontend	Next.js 15 (App Router, TypeScript), Tailwind CSS, shadcn/ui, TipTap, Recharts	One deploy for UI and API routes; TipTap marks suit span highlights	React plus a separate Express server
Worker	Python 3.12, FastAPI, Pydantic, pgmq consumer	OCR, OpenCV, PDF parsing and embeddings are mature in Python	Node workers
Data	Supabase: PostgreSQL 16+, RLS, pgvector, Storage, Auth, Realtime, Queues	Auth, RLS, vectors, storage, realtime and queues in one service	MongoDB plus a separate vector DB
Parsing	PyMuPDF (fitz), python-docx	Span-level text with (page, x0, y0, x1, y1); PyMuPDF is AGPL-3.0, disclosed	pdfplumber, slower on long decks
Vision	OpenCV (ORB features + multi-scale matchTemplate)	Deterministic, explainable boxes with no training data	Training a detector in 14 days
OCR	PaddleOCR with Tesseract fallback	Stronger on stylised ad text; Tesseract is the light fallback	Cloud OCR, which sends images out
Colour / WCAG	scikit-image CIEDE2000 ΔE; WCAG 2.2 relative-luminance contrast	Standard, testable implementations	Hand-rolled colour maths
Embeddings	BAAI/bge-small-en-v1.5 (384-d), run locally in the worker	Runs on CPU, so vault text never leaves the worker	A hosted embedding API
LLM	Claude or Gemini via API, structured JSON output; provider set by config	Structured JSON and vision; the switch avoids lock-in	A self-hosted LLM, too slow on student hardware
Quality / ops	pytest, Vitest, Playwright, golden-set eval in CI, self-hosted Langfuse, GitHub Actions	Precision gate on every change; traces stay inside the trust boundary	Langfuse Cloud, which stores prompts outside our boundary

Postgres Schema & Data Sources

Campaign scope is a first-class column: facts and content both carry `campaign_id`, which is what lets Stage 3 catch context bleed between parallel campaigns. Every finding stores its quote, explanation, rewrite, source and the model and prompt versions that produced it.

Core schema (PostgreSQL)



PK primary key · **FK** foreign key · every table carrying `org_id` has an RLS policy scoped to the caller's memberships. `findings.rule_id` / `fact_id` are nullable; a CHECK requires at least one. `campaigns` and `campaign_id` highlighted in vermilion.

Data sources

- Organisation material: brand guide PDF, Fact Vault (CSV / sheet), glossary, logos, palette, fonts
- WCAG 2.2 AA success criteria for accessibility rules
- Optional claim rule pack based on the ASCI Code
- Golden set: fictional PayWise brand kit + ~40 seeded drafts with labelled issues

Integrations

- **Build window:** file upload, URL import, email and Slack approval notifications
- **Next:** Google Drive import; REST webhook so a CMS can request review before publish
- All third-party services listed on an in-app “Services & AI use” page

Originality: Citation Gate, Intent Verification & Guideline Telemetry

Most entries to this brief will build an LLM that scores drafts. BrandLens treats the LLM as untrusted: facts are verified without it, and its output is shown only after a citation gate. The idea we lead with came from our own interviews: context bleed between parallel campaigns, which none of the tools we compared targets.

[01]

Campaign-scoped Fact Vault

Facts carry a campaign or event scope and an expiry. Stage 3 matches extracted dates, venues and fees against the active scope, which catches LLM context bleed between parallel campaigns.

[02]

Citation gate

Stage 6 accepts a citation only if the rule or fact ID was retrieved for this draft and its dimension matches. Explainability is enforced in code.

[03]

Intent verification

Rewrites are generated in the same LLM call as the finding, then checked locally: bge-small cosine ≥ 0.82, CTA retention and a re-run of Stage 3, in under 15 ms.

[04]

Guideline Clarity Score

Dismissals, reviewer overrides and low-confidence rates combine into a per-rule score; rules below 65 are flagged UNCLEAR.

[05]

Confidence-gated automation

Severity and confidence decide the path. High-confidence minor fixes get a one-click accept by the creator. Blockers and low-confidence findings always go to a person.

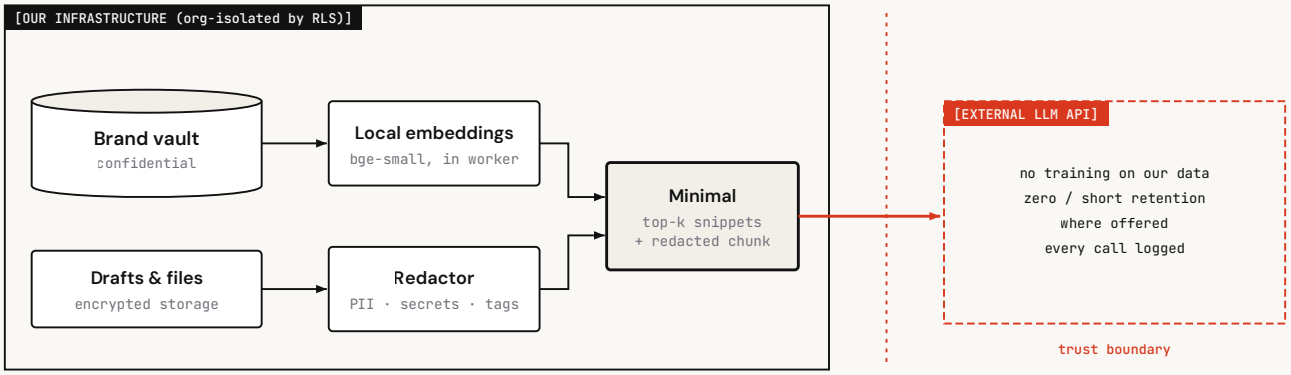
Comparison with existing tool categories

CAPABILITY	GRAMMAR / STYLE ASSISTANTS	ENTERPRISE CONTENT GOVERNANCE	BRAND PORTALS / DAM	GENERAL AI CHAT	BRANDLENS
Tone & terminology checks	●	◐	○	◐	●
Facts & claims verified against approved sources, with citations	○	◐	○	◐	●
Visual brand checks on images	○	○	◐	◐	●
Accessibility checks on content	○	◐	○	◐	●
Review & approval workflow with audit trail	○	◐	◐	○	●
Per-rule clarity telemetry	○	◐	○	○	●
Rewrites verified against stated intent	◐	○	○	◐	●
Within reach of a mid-size team	●	○	◐	●	●
Native integrations with CMS and design tools	●	●	◐	○	◐

● core strength ◐ partial / depends on setup ○ not a focus. Category-level view from public product information (examples: Grammarly Business; Acrolinx; Frontify, Bynder). Individual products vary. Our known weakness is integrations: at launch BrandLens works through upload, URL import and a webhook, not native plug-ins.

Responsible Design: Data Minimisation, RLS Isolation & Human Sign-off

Brand material often includes unreleased products, pricing and legal positions. Parsing, OCR, embeddings and deterministic checks run inside our worker. Only a redacted draft chunk and the retrieved rule snippets are sent to the external LLM, and every consequential action requires a person.



Privacy

- The full vault is never sent to the LLM, only the top-k snippets relevant to the draft
- Embeddings computed locally; vault sections can be marked “never send”
- Org-level retention settings, export and delete; each LLM provider’s retention period listed; designed to align with India’s DPDP Act, 2023

Security

- Supabase Auth with Google sign-in (SAML SSO later); RLS on every org-scoped table
- Signed, expiring Storage URLs; secrets only in server environment
- Agency roles see findings and rule summaries, not confidential source text
- Prompt injection: drafts are passed as delimited data with schema-bound output, so “ignore your rules” in a draft is treated as text; hidden white or tiny text in PDFs is raised as a finding of its own

Human oversight

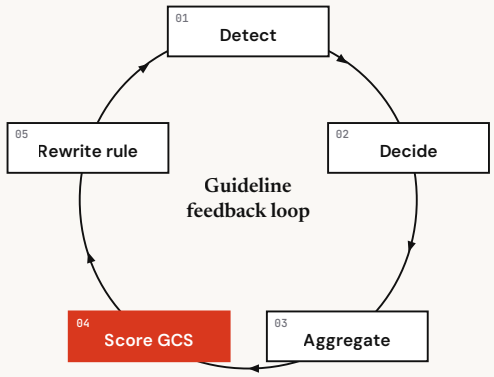
- No auto-publish; no change applied without a user action
- Blockers and findings with confidence < 0.6 require a person
- Overrides allowed, logged with a reason, reviewable in the audit log

Transparency & reliability

- “AI-assisted” label on LLM findings; a Services & AI-use page names each provider and model
- Golden-set evaluation on every prompt or model change, including clean drafts to measure false positives; Langfuse is self-hosted so traces stay inside the trust boundary
- UI built to WCAG 2.2 AA; inclusive-language rules reviewed for bias

Governance Insights: Per-Rule Guideline Clarity Score (GCS)

Every finding, dismissal and approval decision is logged against the rule or fact it cited. Aggregated per rule, these events show which guidelines creators and reviewers apply inconsistently.




```
GCS(r) = 100 × (1 - [
      0.45 × DismissalRate(r)
    + 0.35 × ReviewerOverrideRate(r)
    + 0.20 × LowConfidenceRate(r) ])
```

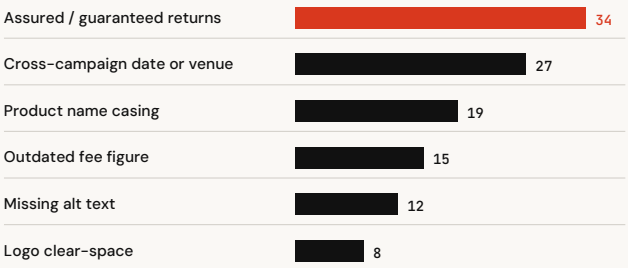
TERM	WEIGHT	COMPUTED AS
DismissalRate	0.45	findings on r dismissed by the creator as "doesn't apply" ÷ findings on r ("AI is wrong" is counted as detector error, not rule ambiguity)
ReviewerOverrideRate	0.35	approvals with an open r-finding, or blocks that cite r on r-clean drafts ÷ reviews
LowConfidenceRate	0.20	r-findings with model confidence < 0.6 ÷ findings on r

A rule with `GCS(r) < 65` after `N ≥ 10` evaluations is flagged **UNCLEAR** and queued for few-shot example generation. The brand owner approves the new examples before they enter retrieval.

Insights dashboard (mock-up, illustrative data)

86	72%	1.4 d	9
avg. brand-fit score	approved on first review	median time to approval	blockers caught this week

Top recurring issues · last 30 days



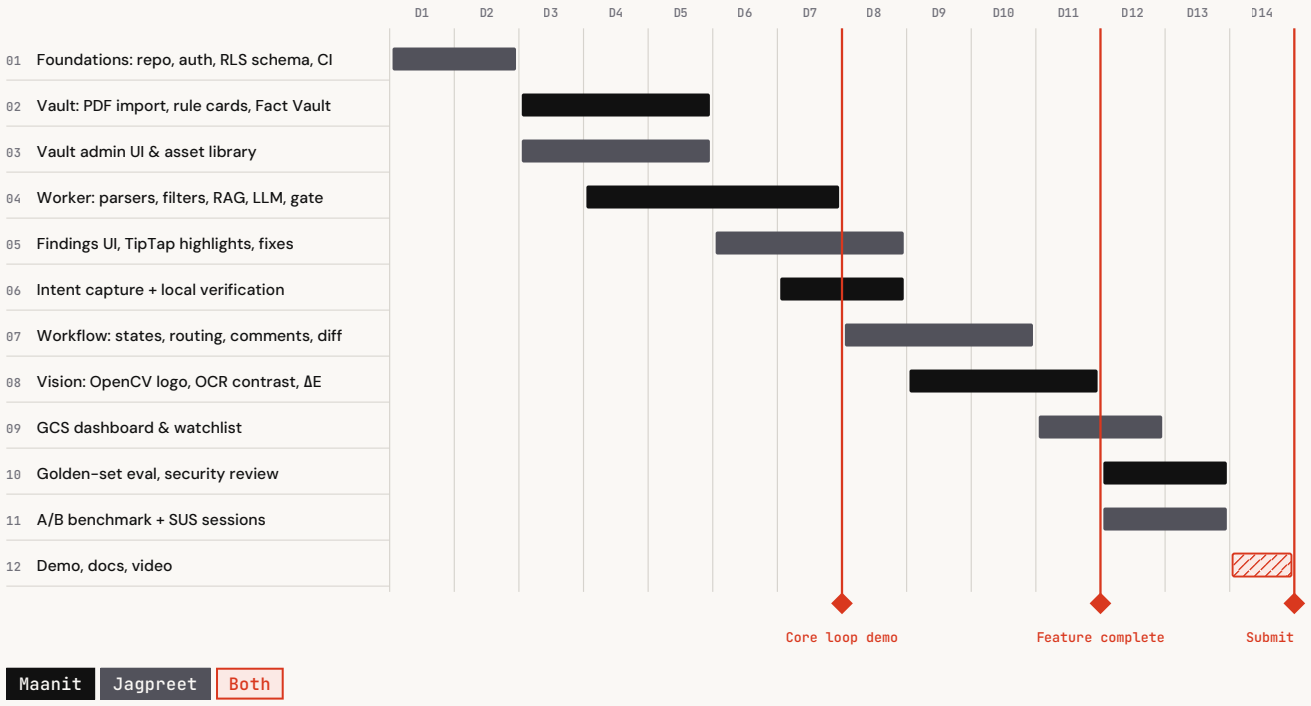
Guideline watchlist

RULE	N	GCS	STATUS
TV-04 "Playful, not flippant"	41	**60**	UNCLEAR
VS-02 Image warmth	22	**73**	watch
CL-03 No assured returns	63	**97**	clear
G-07 PayWise casing	88	**99**	clear

TV-04: 3 do/don't examples generated, pending brand-owner approval. Rules scoring 65–75 are shown as watch.

Build Plan & Role Allocation (14-Day Window)

Planned for a 14-day build window; phases scale proportionally if the official window differs. The text review loop is complete by day 7. No product code exists yet: development starts when the window opens, and the commit history will reflect that.



[A // M. GUPTA]

Maanit Gupta

AI & backend lead

- FastAPI worker: parsers, filters, retrieval, LLM orchestration, citation and intent gate
- Postgres schema, RLS policies, security review
- OpenCV / OCR visual checks
- Golden-set evaluation and reliability metrics

[B // J. SINGH]

Jagpreet Singh

Product, frontend & UX lead

- Next.js client: editor, findings UI, review queue, GCS dashboard
- Workflow states, comments, version diff
- A/B benchmark and SUS usability sessions
- Demo script, video and README

Definition of done at each milestone

Day 7 · Core loop

Import a guide, confirm rules, submit a text draft, receive cited findings with fixes, accept or dismiss, approve. Deployed.

Day 11 · Feature complete

Images and PDFs reviewed with bbox overlays, risk routing, comments and diff, GCS dashboard.

Day 14 · Submit

Golden-set scores, benchmark and SUS results, README with architecture, disclosures and contributions.

Working agreement: daily 15-minute sync, feature branches with pull-request review by the other member, issues tracked on a GitHub project board, and AI-assisted code and templates disclosed in the README as General Rules 6–7 require.

Plan Across All Three Rounds

What each round asks for, what we commit to, and how it is scored. Dates follow the official build window. PENDING · official round dates

R1 · THE PLAN
this document · four interviews

R2 · THE BUILD
working MVP · repo · deployed link

R3 · THE FINISH
final product · demo video · report

ROUND	THE BRIEF REQUIRES	WE COMMIT TO	JUDGED ON (POINTS)
1 · Plan	Problem and evidence, users, research, solution and workflow, features, stack, responsible design, plan and impact	This document and four interviews	Understanding 25, solution 25, technical 20, originality 15, feasibility 15
2 · Build	Working MVP, public repo, deployed link, architecture and key decisions, APIs, models, datasets, libraries, templates and services used, user-testing evidence, each member's contribution	Day 7 core loop deployed, Day 11 feature-complete, three improvements with tests, A/B and SUS results, README disclosures	End-to-end workflow 30, depth and reliability 25, progress vs plan 20, three improvements 15, usability 10
3 · Finish	Final product and repo, demo video or live demo, short report, testing and reliability evidence, changes made, privacy, security and accessibility note, limitations and roadmap, contribution details	Fixes from Round 2 feedback, degraded mode and a load test, a change log, demo video, final report	Completeness 25, three improvements delivered 20, reliability 20, usability and fit 20, docs and roadmap 15

Three improvements committed for Round 2, with pass/fail tests

1

Measured accuracy

Citation gate live; golden-set results published: ≥ 85% precision, ≥ 80% recall on seeded blockers.

PASS IF the eval script is in the repo; ≥ 85% precision and ≥ 80% recall on seeded blockers; zero uncited findings shown.

2

Image & PDF verification

OpenCV logo and clear-space, OCR contrast and ΔE checks; PDF findings with page bounding boxes.

PASS IF the logo is found on at least 9 of 10 test creatives; contrast ratios match WebAIM's checker on 20 samples.

3

GCS telemetry

Per-rule Guideline Clarity Score, UNCLEAR flagging and few-shot example queue.

PASS IF every rule with 10+ evaluations from Round 2 sessions has a score; one UNCLEAR rule is rewritten and shown before and after.

Engineering Feasibility, Risk Matrix & Round 2 Technical Milestones

Mature components

pgvector HNSW, structured LLM output, PyMuPDF, OpenCV, PaddleOCR and CIEDE2000 are well documented. Our work is orchestration, the gate logic and the product.

Vertical slice first

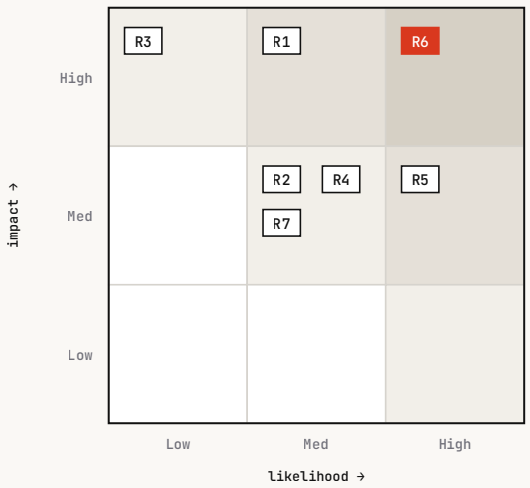
Text review end to end by day 7 gives a working demo early. Images, PDFs and the GCS dashboard are added on top.

Team experience

Maanit has shipped full-stack and ML-integrated apps, including Supabase and LLM API builds at previous hackathons. Roles are split so each member owns one tier.

PENDING · Jagpreet's shipped projects

Risk heat map



#	RISK	MITIGATION
R1	False-positive findings	Citation gate, confidence routing, golden-set precision target ≥ 85%
R2	Latency above SLA	Deterministic stages first, one LLM call per draft, cached retrieval
R3	Confidential data exposure	Redaction, top-k context only, no-training API settings, RLS
R4	Rule extraction errors from PDF	AI drafts rule cards; admin confirms each one
R5	Warning fatigue for creators	Severity thresholds; GCS flags noisy rules for rewrite
R6	Scope vs. two-person team	MoSCoW order; text loop complete by day 7; cut-lines below
R7	Logo / contrast false matches	Multi-scale ORB + template thresholds tuned on golden set; vision LLM only suggests

Cut-lines & Plan B

If Day 7 slips, drop in this order

- 1 · URL import (upload stays)
- 2 · ΔE palette check
- 3 · Logo clear-space check
- 4 · Version diff view (keep version history)
- 5 · Slack notifications (keep email)

Never cut: the claims and facts loop, citation gate, approval states and audit log.

Plan B per risky part

- PaddleOCR too heavy on the host → Tesseract only
- ORB matching unreliable → vision-LLM suggestion, labelled as a suggestion
- LLM timeout → deterministic findings return with a banner; retry with backoff; idempotent jobs
- SUS recruitment short → minimum 8 participants
- Long PDFs → chunk by page, one LLM call per chunk, capped at 40 pages

Validation: Phase 1 Discovery Results & Phase 2 Benchmark Plan

≥ 85%	100%	< 15 s	≥ 70
finding precision on the golden set	of displayed findings carry a valid citation	P95 text analysis (≤ 2,000 words)	System Usability Scale score

✓ PHASE 1 · COMPLETED (ROUND 1)

Primary discovery interviews across 4 distinct content environments (agency, university cultural/service orgs, SMB content creation)

INTERVIEWEE	KEY FINDING	REQUIREMENT DERIVED
Pallavi Biala Leo Club, Panjab University	3–4 approval rounds; LLM context bleed put the wrong venue and time into a post	Deterministic Fact Vault match on dates, venues, campaign facts
Harshit Singh Raghuvanshi Agency founder, Mohali	3-stage chain; work made for the creator, not the viewer; late founder rewrites	Mandatory audience/intent at submission; risk-sorted queue with diffs
Shaurya Sandal Cultural Society, Symbiosis	Late, vague, contradictory feedback under deadline pressure	Rule-cited fixes before submit; rewrites keep the campaign hook
Devansh SMB content creator	Static PDF guide; no data on which rules confuse creators	PDF → rule cards; per-rule Guideline Clarity Score

PHASE 2 · ROUND 2 BUILD WINDOW

Controlled A/B review benchmark (n = 10 seeded drafts) & SUS usability testing (n = 15 participants)

A/B benchmark. Reviewers check 10 drafts seeded with labelled issues (claims, fact drift, terms, contrast) in a crossover design: each reviewer gets 5 drafts with BrandLens findings and 5 without, swapped between reviewers, so no one sees a draft twice.

Measured: issues caught (recall), false flags accepted, minutes per draft.

SUS testing. 15 participants complete submit → resolve findings → approve, recruited from the agency and student-org networks reached in Phase 1.

Measured: SUS score, task completion, time on task, friction points.

Expected impact, with baselines and how each metric is measured, is on page 24.

Expected Impact: Metrics, Baselines & Measurement

Baselines come from our four interviews; targets are measured in the Round 2 A/B benchmark, the golden set and the product's own audit log.

72–96 h	≈ 2 h	3–4 → 1–2	≥ 80%
TURNAROUND TODAY	TARGET TURNAROUND	REVIEW ROUNDS	BLOCKERS CAUGHT EARLY
per post, reported in interviews	pre-check + one risk-routed sign-off	per post	before human review, golden set

Metrics by user

WHO	METRIC	BASELINE	TARGET	MEASURED BY
Approver	Turnaround per post	72–96 h (interviews)	**About 2 h**	Round 2 A/B timing
Team	Review rounds per post	3–4 (interviews)	**1–2**	A/B test and audit log
Creator	Seeded blockers caught before human review	No systematic fact check today	**≥ 80% recall**	Golden set
Legal	Overrides without a logged reason	Not tracked today	**Zero**	Audit log
Readers	Drafts checked for contrast and alt text	Not checked today	**All drafts**	Findings log
Brand owner	Unclear rules identified	None today	**Flagged within two weeks of use**	GCS watchlist

Where the impact lands

Organisation impact

Fewer post-publication corrections, shorter approval chains, an audit trail for regulators and clients.

Team impact

Creators get specific, cited feedback; approvers review risk and intent instead of every line.

Public impact

Fewer unsupported claims reach consumers; contrast and alt text checked by default.

Future Scope After the Event

The roadmap extends the same verification core to more channels, languages and regulators, then moves upstream so content is checked before a person ever sees it.

H1 · FIRST MONTHS AFTER THE EVENT

- Pilot with our interviewees' teams, if they agree
- Google Drive import
- WordPress or Webflow webhook
- Slack approvals

H2 · WITHIN A YEAR

- Google Docs, Canva and Figma add-ins
- Chrome extension
- Hindi, Tamil and Hinglish tone checks
- Rule packs for ASCI, SEBI, IRDAI and FSSAI advertising rules
- Captions and transcripts
- Self-hosted LLM option

H3 · LONGER TERM

- Checking live channels after publication
- An API so AI writing tools check copy before a person sees it

Business model

Free

STUDENT CLUBS & NGOS

where three of our four interviewees work

Per seat

AGENCIES

priced per reviewing seat

Self-hosted

ENTERPRISES

for teams that keep brand material on their own infrastructure

Team & Participant Details

Team name	PENDING · exactly as on the form
Product	BrandLens
Problem statement	Inkloom AI — Intelligent brand-governance platform
Institution	SRM Institute of Science and Technology
Point of contact	PENDING · name, email, phone

Participants

FULL NAME (AS ON CERTIFICATE)	ROLE	DEPARTMENT & YEAR	REGISTER NO.	EMAIL	PHONE	GITHUB
Maanit Gupta	AI & backend lead	B.Tech CSE, 2nd year	PENDING · reg. no.	PENDING · email	PENDING · phone	maanit-gupta
Jagpreet Singh	Product, frontend & UX lead	PENDING · dept, year	PENDING · reg. no.	PENDING · email	PENDING · phone	PENDING · GitHub

Individual contributions

[A // M. GUPTA]

Maanit Gupta

AI & backend lead

OWNS END TO END

FastAPI verification worker: parsers, filters, retrieval, LLM orchestration, citation and intent gate; Postgres schema and RLS policies; OpenCV / OCR checks; golden-set evaluation.

ROUND 1

PENDING · Round 1 contribution

[B // J. SINGH]

Jagpreet Singh

Product, frontend & UX lead

OWNS END TO END

Next.js client: editor, findings UI, review queue and GCS dashboard; workflow states, comments and version diff; A/B benchmark and SUS sessions; demo, video and README.

ROUND 1

PENDING · Round 1 contribution

Both members will describe their own role on the submission form in the same words used here. Contributions will be visible in the repository's commit and review history.

References & Disclosures

1 **Lucidpress (now Marq),** *The State of Brand Consistency*, **2019**

2 **Marq (formerly Lucidpress),** *Content Effectiveness Report* **and** *The State of Brand Consistency*, **2021**
https://pub.lucidpress.com/brand-consistency/

3 **Advertising Standards Council of India (ASCI),** *Annual Complaints Report 2024–25*, **press release, 28 May 2025**
https://www.ascionline.in/wp-content/uploads/2025/05/ASCI-Press-Release-Annual-Complaints-Report-2024-25-1.pdf

4 **ASCI,** *Half-Yearly Complaints Report 2025–26* **(April–September 2025)**
https://www.ascionline.in/wp-content/uploads/2025/11/ASCI-Half-Yearly-Complaints-Report-2025-26.pdf

5 **Kaltura,** *Riding the wave: a 2024 survey report on the use of AI technologies in marketing teams* **(n=500; US, UK, France, Germany, Italy; Aug–Sep 2023), as reported by MediaPost**
https://www.mediapost.com/publications/article/390709/

6 **Boston Consulting Group, CMO survey on GenAI (n=200, April 2024), press release 12 June 2024**
https://www.bcg.com/press/12june2024-cmos-cite-genai-adoption-top-five-priority-over-next-year

7 **WebAIM,** *The WebAIM Million: 2025 report*
https://webaim.org/projects/million/2025

8 **W3C,** *Web Content Accessibility Guidelines (WCAG) 2.2*
https://www.w3.org/TR/WCAG22/

Statistics are paraphrased from the cited reports and were checked against them in October 2026.

Disclosures

Authorship & AI assistance (General Rule 7)

Primary user research (interviews with marketing leads and creators in Chandigarh, Mohali and Pune), product decisions and the final system specification are the work of Maanit Gupta and Jagpreet Singh. Claude (Anthropic) assisted with desk-research synthesis, drafting of text and architecture options, and visual layout and diagram rendering. The team reviewed, revised and approved all content.

Pre-existing work

No BrandLens product code was written before the build window. Open-source libraries listed on page 15 will be used under their licences; PyMuPDF is AGPL-3.0, which is compatible with our public repository and will be disclosed with every other licence in the README. Product names in the comparison are trademarks of their owners; the demo brand “PayWise” and its facts are fictional.

BrandLens

Maanit Gupta · Jagpreet Singh · SRM Institute of Science and Technology